# Bayesian Filtering Approaches for 3D Localization of Quadruped Robots on the Grand Tour Dataset

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## Introduction

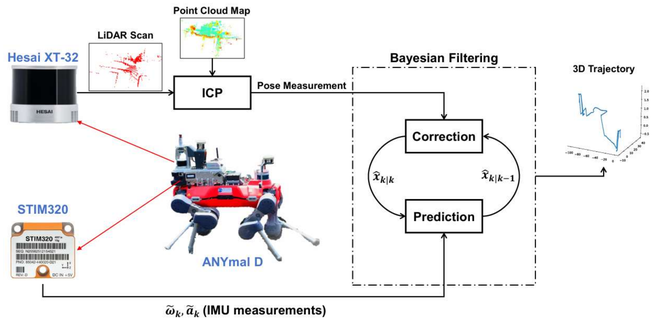
### Motivation

- Accurate state estimation is a cornerstone challenge for legged platforms operating in complex environments due to terrain irregularities, foot slipping, and sensor noise.
- Need for a systematic evaluation of state estimation algorithms to determine the most robust strategy for specific constraints and environmental conditions.

### Objectives

- Conduct a comprehensive benchmark of mainstream Bayesian filtering frameworks using the ANYmal GrandTour Dataset.
- Deeply investigate the impact of different approximation and sampling strategies on 3D localization performance in real-world scenarios.

## Methodology




### 1 IMU-LiDAR State Estimation via Bayesian Filtering

- **Frameworks:** Build the map, and then fuse IMU prediction with LiDAR ICP scan-to-map correction and compare four frameworks—EKF (first-order linearization), UKF (sigma-point nonlinear handling), PF (non-Gaussian, multi-modal), and InEKF (Lie-group-consistent error dynamics).

## 2 Algorithm Implementation

**Algorithm 1** PF-Based Radar–IMU Trajectory Fusion

**Require:** Map $\mathscr{M}$, IMU $\mathscr{D}_{\text{imu}}$, LiDAR $\mathscr{D}_{\text{lidar}}$, Extrinsic $\mathbf{T}_{\text{IL}}$
**Ensure:** Trajectory $\mathscr{T}$
Init: Load map $\mathscr{M}$; Align 1st scan via ICP; Init particles $\{\mathbf{x}_i, w_i\}_{i=1}^{N}$;
**while** New LiDAR frame $\mathbf{P}_k$ arrives **do**
  **1. Prediction (IMU Propagation):**
  **for** each particle $i$ **do**
    Integrate IMU $\mathscr{D}_{\text{imu}}$ to update $(\mathbf{R}_i, \mathbf{v}_i, \mathbf{p}_i)$;
    Add process noise $\mathbf{\varepsilon}_i$;
  **end for**
  **2. Multi-Candidate ICP:**
  Select $K$ indices $\mathscr{I}$ (Best, Mean, Random);
  **for** each $j \in \mathscr{I}$ **do**
    $\mathbf{T}_{\text{init}} \leftarrow \mathbf{T}_{\text{WI}}^{(j)} \mathbf{T}_{\text{IL}}$;
    $\mathbf{T}_{\text{ICP}}^{(j)} \leftarrow \text{ICP}(\mathbf{P}_k, \mathscr{M}, \mathbf{T}_{\text{init}})$;
    $s_j \leftarrow \text{Fitness} - \lambda \cdot \text{RMSE}$;
  **end for**
  $(j^*, \mathbf{T}^*) \leftarrow \arg\max_j (s_j)$;
  **3. Update & Gating:**
  **if** Fitness$(\mathbf{T}^*) > \tau_f$ and RMSE$(\mathbf{T}^*) < \tau_r$ **then**
    **for** each particle $i$ **do**
      Update weight $w_i \propto w_i \cdot \mathscr{N}(\mathbf{x}_i; \mathbf{T}^*, \Sigma)$;
    **end for**
    Normalize $w_i$;
  **end if**
  **4. Estimation & Resample:**
  Estimate $\hat{\mathbf{x}} \leftarrow \sum \mathbf{w}_i \mathbf{x}_i$;
  **if** $N_{\text{eff}} < N_{\text{th}}$ **then**
    Resample $\{\mathbf{x}_i\}$; Apply Roughening;
  **end if**
  Store $\hat{\mathbf{x}}$ in $\mathscr{T}$;
**end while**
**return** $\mathscr{T}$

**Algorithm 1** UKF-Based Radar–IMU Trajectory Fusion

**Require:** previous state mean $\mu_{k-1}$, previous covariance $\Sigma_{k-1}$, IMU angular velocity $\omega_k$, IMU linear acceleration $a_k$, radar position measurement $z_k$
Estimate gyro bias, initial gravity direction, and residual acceleration bias from the initial IMU segment
Integrate $\omega_k$ to update the sensor orientation
Transform $a_k$ from the body frame to the world frame and remove gravity
$\mathscr{X}_{k-1} \leftarrow$ generate sigma points from $\mu_{k-1}$ and $\Sigma_{k-1}$
$\mathscr{X}_k^- \leftarrow$ compute the corresponding sigma-point weights
**for** each sigma point $\mathscr{X}_{k-1}^{(i)}$ **do**
  $\mathbf{x}_k^{(i)} \leftarrow f(\mathscr{X}_{k-1}^{(i)}, a_k, \Delta t)$
**end for**
$\mu_k^- \leftarrow \sum_{i=0}^{2n} w_i^{(i)} \mathbf{x}_k^{(i)}$
$\Sigma_k^- \leftarrow \sum_{i=0}^{2n} w_i^{(i)} (\mathbf{x}_k^{(i)} - \mu_k^-)(\mathbf{x}_k^{(i)} - \mu_k^-)^T + Q_k$
Transform the radar ICP trajectory into the IMU frame using the static extrinsic calibration
$z_k \leftarrow [p_x, p_y, p_z]^T$
$\mathscr{Z}_k^- \leftarrow$ generate sigma points from $\mu_k^-$ and $\Sigma_k^-$
$w \leftarrow$ compute the corresponding sigma-point weights
**for** each predicted sigma point $\mathscr{Z}_k^{-(i)}$ **do**
  $\mathbf{z}_k^{(i)} \leftarrow h(\mathscr{Z}_k^{-(i)})$
**end for**
$\hat{z}_k \leftarrow \sum_{i=0}^{2n} w_i \mathbf{z}_k^{(i)}$
$y_k \leftarrow z_k - \hat{z}_k$
$S_k \leftarrow \sum_{i=0}^{2n} w_i (\mathbf{z}_k^{(i)} - \hat{z}_k)(\mathbf{z}_k^{(i)} - \hat{z}_k)^T + R_k$
$\Sigma_{xz} \leftarrow \sum_{i=0}^{2n} w_i (\mathscr{Z}_k^{-(i)} - \mu_k^-)(\mathbf{z}_k^{(i)} - \hat{z}_k)^T$
$K_k \leftarrow \Sigma_{xz} S_k^{-1}$
$\mu_k \leftarrow \mu_k^- + K_k y_k$
$\Sigma_k \leftarrow \Sigma_k^- - K_k S_k K_k^T$
**return** $\mu_k, \Sigma_k$

**Algorithm 1** EKF Algorithm

**Require:** previous state mean $\mu_{k-1}$, previous covariance $\Sigma_{k-1}$, control input $u_k$, measurement $z_k$, process noise covariance $Q_k$, measurement noise covariance $R_k$, nonlinear state transition $f(\cdot)$, nonlinear measurement $h(\cdot)$
**// Prediction (Time Update)**
Predict the state mean
$\bar{\mu}_k^- \leftarrow f(\mu_{k-1}, u_k)$
Compute the state transition Jacobian
$F_k \leftarrow \frac{\partial f(x,u)}{\partial x}\Big|_{x=\mu_{k-1}, u=u_k}$
Predict the covariance
$\bar{\Sigma}_k^- \leftarrow F_k \Sigma_{k-1} F_k^T + Q_k$
**// Update (Measurement Update)**
Predict the measurement
$\hat{z}_k \leftarrow h(\bar{\mu}_k^-)$
Compute the measurement Jacobian
$H_k \leftarrow \frac{\partial h(x)}{\partial x}\Big|_{x=\bar{\mu}_k^-}$
Compute the innovation (residual)
$v_k \leftarrow z_k - \hat{z}_k$
Compute the innovation covariance
$S_k \leftarrow H_k \bar{\Sigma}_k^- H_k^T + R_k$
Compute the Kalman gain
$K_k \leftarrow \bar{\Sigma}_k^- H_k^T S_k^{-1}$
Update the state mean
$\mu_k \leftarrow \bar{\mu}_k^- + K_k v_k$
Update the covariance
$\Sigma_k \leftarrow (I - K_k H_k)\bar{\Sigma}_k^- (I - K_k H_k)^T + K_k R_k K_k^T$
*or* $\Sigma_k \leftarrow (I - K_k H_k)\bar{\Sigma}_k^-$
**return** $\mu_k, \Sigma_k$

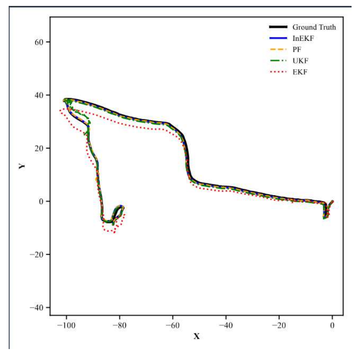
**Algorithm 1** RIEKF (Right-Invariant Error EKF)

**Require:** previous estimate $R_{k-1}, p_{k-1}, v_{k-1}, b_{g,k-1}, b_{a,k-1}$, previous covariance $P_{k-1}$, IMU angular velocity $\omega_k$, IMU linear acceleration $a_k$, measurement $z_k$ (e.g., radar position), time step $\Delta t$
Estimate gyro bias, initial gravity direction, and residual acceleration bias from the initial IMU segment
$\hat{\omega}_k \leftarrow \omega_k - b_{g,k-1}$, $\hat{a}_k \leftarrow a_k - b_{a,k-1}$
$R_k \leftarrow R_{k-1} \exp\left((\hat{\omega}_k)^\wedge \Delta t\right)$
$a_k^w \leftarrow R_k \hat{a}_k - g$
$v_k \leftarrow v_{k-1} + a_k^w \Delta t$
$p_k \leftarrow p_{k-1} + v_{k-1} \Delta t + \frac{1}{2} a_k^w \Delta t^2$
$\Phi_k, Q_k \leftarrow$ from RIEKF error dynamics
$P_k \leftarrow \Phi_k P_{k-1} \Phi_k^T + Q_k$
$p_k^- \leftarrow R_k p_k + p_r$
$\hat{z}_k \leftarrow z_k - p_k^-$
$H_k \leftarrow \frac{\partial h(\zeta)}{\partial \zeta}\Big|_{\zeta_k^-}$
$S_k \leftarrow H_k P_k H_k^T + R_k$
$K_k \leftarrow P_k H_k^T S_k^{-1}$
$\delta\xi_k^R \leftarrow K_k \hat{z}_k$
$R_k \leftarrow R_k^- \exp\left((\delta\theta_k)^\wedge\right)$
$p_k \leftarrow p_k^- + \delta p_k$
$v_k \leftarrow v_k^- + \delta v_k$
$b_{g,k} \leftarrow b_{g,k}^- + \delta b_{g,k}$
$b_{a,k} \leftarrow b_{a,k}^- + \delta b_{a,k}$
$P_k \leftarrow (I - K_k H_k) P_k (I - K_k H_k)^T + K_k R_k K_k^T$
**return** $R_k, p_k, v_k, b_{g,k}, b_{a,k}, P_k$

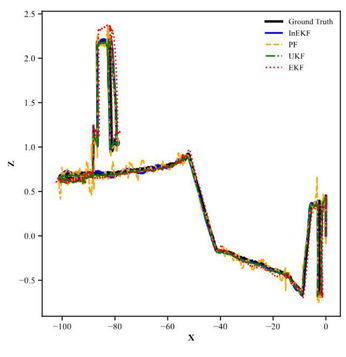
## Results

- We evaluate our method on three representative datasets (ETH-1, SPX-1, and SNOW-1) covering diverse environments, and assess localization accuracy using Absolute Trajectory Error (ATE). Here we only show the result from ETH-1 as example.
- The overall performance is quantified by the RMSE of ATE, measuring the deviation between estimated and ground truth trajectories over time.

### 1 Trajectory Comparison

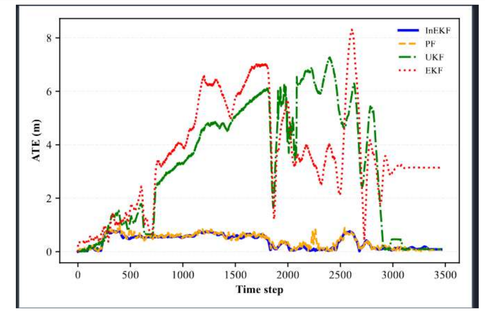


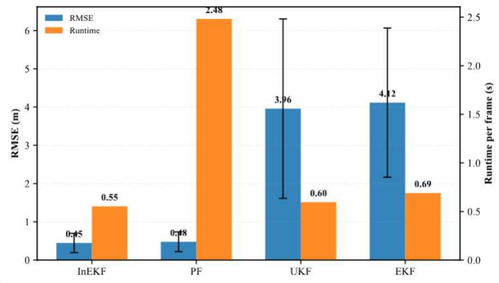

Top view




Side view

## 2 Error and Runtime comparison




## Discussion

- Our approach achieves high performance with real-time capability, especially regarding the In-EKF filter.
- All methods perform similarly in simple environments, but InEKF shows better robustness in challenging scenarios, while PF is more computationally expensive.
- The main limitation is the reliance on ICP measurements, which can lead to drift when the measurements become unreliable.

## Future Works

- Future work will focus on improving robustness under perceptual degeneracy.
- Another promising direction is to incorporate contact information within an invariant filtering framework.

**References:** [1] J. Frey, T. Tuna, F. Fu, K. Patterson, T. Xu, M. Fallon, C. Cadena, and M. Hutter,``GrandTour: A Legged Robotics Dataset in the Wild for Multi-Modal Perception and State Estimation,"arXiv:2602.18164, 2026.
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